

gmanifold: sampling results

TL;DR

Fit one decoder $G: R^{16} \rightarrow R^D$ per residual location on the real token clouds of SimpleStories-5M, -35M and TinyLlama-1.1B (prefix + every real token; also real story occurrences), then sample on the learned surface within $\alpha \times$ spacing of the observed states and validate against three independent checks with calibration bands.

- Samples are on-support for $\alpha \leq 0.7$ at every location of all three models (kernel score above held-out real states, tangent residual and nearest-real distance well inside the noise bands, coverage of the real cloud 0.97–0.99); $\alpha = 1$ is marginal, $\alpha = 1.5$ is off-support. Results are stable over seeds, latent dimension and prefixes.
- Pushed through the real model, the samples' images are scored like images of real held-out states by independently fitted destination manifolds.
- The sampler is uniform *within the tube around the observed states* (anchors equalised, clearly not the empirical token density), not in the manifold's own volume; the per-anchor rule that would fill sparse regions collapses.
- The fit is limited by the latent dimension, not by the network or the optimiser (63-fit capacity study); a VAE would be worse for this purpose.
- Two robustness rules matter: drop degenerate and isolated states first, and use the global support radius.
- Negative: a 32k-token embedding matrix in 2048 dimensions is not chartable by one decoder.
- To the model itself, samples behave like blends of their nearest real tokens: their next-token distributions are closer to the nearest token's than real tokens are to each other (JS 0.21 vs 0.46 at $\alpha = 0.3$; random-token baseline 0.57), with top-1 agreement 32 % (random 6 %); one-spacing noise loses this (JS 0.51, agreement 8 %). Layer by layer, the images stay nearer to real states than held-out real images do and between the images of their two anchors.

Summary

Setup. GlobalManifold(latent_dim=16) fitted per residual location on "prefix + every real token" clouds (SimpleStories-5M: 13 locations, 3 seeds, $m \in \{8, 16, 32\}$; SimpleStories-35M: 25 locations, 5 seeds; TinyLlama-1.1B: 8 locations with 2 seeds and all 45 locations with one seed, 31,885 tokens after dropping 111 untrained byte-fallback rows and 1 massive-activation outlier; plus a second prefix for the 5M model). Samples: 2000 per setting from 16,000 candidates, default radius="global".

What holds everywhere (all models, seeds, prefixes).

1. *On-support regime.* Samples score above held-out real states (kernel $u \geq 1.03$ at every location, coverage of the real cloud 0.97–0.99) for $\alpha \leq 0.7$; at $\alpha = 1.0$ they still sit above the half-spacing noise band (u 0.96 vs 0.91) with coverage 0.93; $\alpha = 1.5$ is off-support (u 0.72–0.74, coverage 0.12). Seed spread of u at $\alpha = 0.3$ is ≤ 0.013 . The independent tangent-chart residual agrees: samples 0.55–0.64 spacings from the nearest local chart for $\alpha \leq 0.7$ (held-out real 0.82, half-spacing noise 0.95, one-spacing noise 1.24). The conservative tangent-chart sampler used as a cross-check gives u 0.97 and nearest-real 0.27 at $\alpha = 0.3$: on-support but, by construction, hugging the anchors.
2. *Latent dimension.* $m = 8 / 16 / 32$ give held-out reconstruction 0.78 / 0.74 / 0.67 spacings and u 1.16 / 1.11 / 1.06 at $\alpha = 0.3$; larger m fits better but the volume weights spread more (ESS 5.5k / 3.4k / 1.5k of 16k).
3. *Volume reweighting works and is validated exactly* on a synthetic sheet with a known volume element (tests/test_manifold.py): with per-anchor radii the samples reproduce the volume-uniform target; with the global radius they lie between the data density and the target at small α and converge to the target as α grows.
4. *Global radius is the robust default.* With per-anchor radii (radius="local") the union of balls is dominated by the sparsest anchors (ball volume $\propto p^m$): on SimpleStories-5M L5.ffn the anchor participation ratio drops from 3546 to ~ 100 and coverage from 0.995 to 0.96; on TinyLlama a single massive-activation state took 100 % of the anchor mass and 99 % of the importance weight (ESS = 1), collapsing every sample onto it. Dropping isolated states (outlier_mask) and using the global radius removes the failure.
5. *Propagation.* Embedding samples pushed through the real model (embed $\rightarrow$ L0.attn / L0.ffn / mid / last) are scored by independently fitted destination manifolds like images of real held-out states (u within ± 0.03 of the real images at every destination for $\alpha \leq 0.5$, ± 0.2 at the first sub-layer where the map is near-identity and the samples' own overshoot carries over). Real states displaced by one spacing before propagation are healed by the network: u 0.71 after one sub-layer, ≥ 0.99 after six or more.
6. *Loss ablation* (5M, embed and L5.ffn, $\alpha = 0.3$). Removing the local-distance term lowers u by 0.04–0.07 and raises the samples' reconstruction gap by 25–40 %; 3× longer training overfits (train recon 0.51 / 0.40 vs 0.67 / 0.52, held-out recon worse, sample ESS 2.0k / 2.1k vs 4.9k / 6.8k, reconstruction gap $\times 3$). Hidden width and the curvature term are neutral.

7. *Real-occurrence clouds* (20k random token occurrences from 2,500 stories, split by story; all contextual locations of SimpleStories-5M and 35M; §6 of the tables). Lower and token-clustered dimension (TwoNN 5–12), harder fits (held-out reconstruction 0.76–1.17 spacings vs 0.70–0.87 for prefix clouds) and softer calibration bands (one-spacing noise scores u 0.86 instead of 0.69). Samples nevertheless stay on-support at every location of both models for $\alpha \leq 1$ ($u \geq 1.01$, above the $0.5\times$ band), and their FFN images are scored by the independent destination fits like real held-out images (u 1.01–1.03 vs 1.00). Coverage is lower (0.65 at $\alpha = 0.3$) only because 2000 samples are spread over 18k states.
8. *Decoder capacity study* (§7 of the tables; 63 fits on embed, L2. ffn, L5. ffn). At $m = 16$ no architecture or training variant changes the held-out reconstruction by more than 0.006 spacings: widths from (256, 128) to (2048, 1024), one to three hidden layers, learning rates $1e-3$ – $5e-3$, batches 128–2048, λ_{geom} 0.03–0.3 all give 0.78–0.80 (embed), 0.72–0.74 (L2. ffn), 0.70–0.72 (L5. ffn); the two largest nets are worse (the 3.7M-parameter three-layer net under-fits in 300 epochs), and 600 epochs only lowers the training error (best validation epoch ≈ 300). The fit is therefore neither capacity- nor optimisation-limited. The one lever is the latent dimension: held-out reconstruction improves smoothly and without plateau from $m = 8$ to 64 (0.83 $\rightarrow$ 0.68, 0.77 $\rightarrow$ 0.57, 0.77 $\rightarrow$ 0.49), which says the clouds have a long spectral tail rather than a sharp intrinsic dimension (local PCA needs 21–27 components for 90–95 % of the neighbourhood variance). Larger m costs sampling efficiency: ESS at $\alpha = 0.3$ drops from 4.9k ($m = 8$) to 25–1200 ($m = 64$) and coverage from 0.99 to 0.81–0.90, while u moves from 1.15 toward 1.03 (samples closer to the real states' own score). $m = 16$ –24 is the practical compromise; $m = 32$ with more candidates if tighter reconstruction matters more than efficiency.
9. *Why not a VAE* (experiments/vae_check.py, results/vae_check.json; same architecture with a Gaussian encoder and β -KL, embed and L5. ffn). With $\beta \leq 1e-3$ the posterior collapses to a point ($\sigma \approx 0.01$ –0.05), reconstruction equals ours, and decoding prior samples $z \sim N(0, I)$ fails (u 0.05–0.16, coverage ≤ 0.15) because the aggregate posterior is nowhere near the prior; our Ω sampler on the posterior means works exactly as before. With $\beta = 1e-2$ the prior sampler becomes usable (u 1.2–1.3, coverage 0.98) at the price of reconstruction (0.79 $\rightarrow$ 0.86 at embed, 0.71 $\rightarrow$ 0.80 at L5. ffn), of latent dimension (9 of 16 units stay active at embed) and of our sampler's ESS (415 / 126: the KL fights the local-distance term). The prior sampler also targets the aggregate-posterior ($\approx$ data) density, not manifold volume, and has no notion of "near the observed support". Kept: nothing; noted as useful by-products: the number of active units as a data-driven m estimate, and the posterior σ as a per-point latent scale.
10. *What "volume-uniform" means on the real embedding cloud* (3940 SimpleStories-5M tokens, $m = 16$, 40k samples). Rank correlation between samples-per-real-anchor and the anchor's cell volume ($\propto r_i^m$): default sampler +0.005 ($\alpha = 0.5$) / -0.004 ($\alpha = 1$), data-density sampler -0.275 , per-anchor-radius sampler +0.015 with 90 % of its samples on 1 % of the anchors and off-support (u 0.57). So the default sampler is uniform over the uniform-thickness tube Ω around the observed states (anchors equalised; clearly not the data density), but not uniform in the manifold's own volume, which would need the large cells of sparse regions to be filled — exactly what the per-anchor rule attempts and fails at. On the synthetic sheet the same effect appears as the global radius sitting between the data density and the exact volume target at small α .
11. *How the samples look to the model* (§8 of the tables, experiments/downstream.py). With an embedding sample at the last position of the prefix and the full 5M model run: the sample's next-token distribution is closer to that of its nearest real token (JS 0.21 at $\alpha = 0.3$, 0.27 at 0.7, 0.37 at 1.5) than real tokens are to their own nearest neighbour (0.46) and far from a random token's (0.57); top-1 predictions agree with the nearest token's in 32 % of the samples (real-to-neighbour 19 %, random 6 %); entropy is slightly higher (3.4 vs 2.9 nats), as expected for a blend. Real tokens displaced by one spacing lose the neighbour relation (JS 0.51, top-1 agreement 8 %, entropy 3.9). Along the network the images of $\alpha \leq 0.7$ samples stay closer to real states than held-out real images do at every location (0.64–0.91 vs 0.89–0.96 spacings), keep $u \approx 1$, and sit 0.5–0.9 spacings from the segment joining the images of their two nearest anchors (real 0.8–0.95, noise 1.1–1.3): they remain interpolations of their neighbours.

What does not hold. The TinyLlama *embedding* cloud (31,885 rows in $D = 2048$, TwoNN ≈ 205) is not chartable by one global decoder: held-out reconstruction stays at 6 spacings for $m = 16$ and $m = 64$ and samples land 2.6 spacings from real rows. Its *contextual* clouds are chartable (TwoNN 8–18, reconstruction 0.66–0.93 spacings) and sample normally at all 44 of them ($u \geq 1.13$ at $\alpha = 0.3$, coverage ≥ 0.90), with lower ESS (≈ 100 –2400) because the volume element varies more in 2048 dimensions. Use *intrinsic_dimension* and the held-out reconstruction before trusting samples on a new cloud.

Point clouds: a fixed prefix followed by every real token, residual state at the last position, one cloud per residual location; 90 % of the tokens fit the manifold, 10 % are held out. All distances are in units of the local spacing (distance to the 8-th real neighbour); u is the Guidotti kernel score (1 on fitted states); bands = held-out real states and real states displaced by $0.5\times$ / $1\times$ their spacing. Samples: 2000 per setting, $8\times$ candidates, volume reweighting.

0. Runs

run	model	D	locations	states	time
35M	SimpleStories/SimpleStories-35M	512	25	3964	5 min
35M_seeds	SimpleStories/SimpleStories-35M	512	25	3964	8 min
5M	SimpleStories/SimpleStories-5M	256	13	3940	17 min
5M_prefix2	SimpleStories/SimpleStories-5M	256	13	3940	6 min
TinyLlama	TinyLlama/TinyLlama-1.1B-Chat-v1.0	2048	8	31885	15 min
TinyLlama_all	TinyLlama/TinyLlama-1.1B-Chat-v1.0	2048	45	31885	34 min
TinyLlama_m64	TinyLlama/TinyLlama-1.1B-Chat-v1.0	2048	5	31885	8 min

1. Intrinsic dimension and fit quality per location

35M (SimpleStories/SimpleStories-35M)

location	TwoNN	MLE	val recon m=16
embed	23.000	28.573	0.867 (rank 16)
L0.attn	23.855	28.275	0.889 (rank 16)
L0.ffn	15.243	27.655	0.801 (rank 16)
L1.attn	15.390	27.768	0.806 (rank 16)
L1.ffn	12.394	24.061	0.779 (rank 16)
L2.attn	12.448	23.696	0.782 (rank 16)
L2.ffn	11.188	20.273	0.768 (rank 16)
L3.attn	10.997	20.091	0.765 (rank 16)
L3.ffn	10.393	19.375	0.756 (rank 16)
L4.attn	10.332	19.209	0.753 (rank 16)
L4.ffn	10.035	18.590	0.743 (rank 16)
L5.attn	9.900	18.511	0.741 (rank 16)
L5.ffn	9.898	17.878	0.734 (rank 16)
L6.attn	9.870	17.718	0.730 (rank 16)
L6.ffn	10.133	17.067	0.724 (rank 16)
L7.attn	9.956	16.577	0.721 (rank 16)
L7.ffn	9.941	16.389	0.715 (rank 16)
L8.attn	9.989	16.329	0.713 (rank 16)
L8.ffn	9.950	16.504	0.714 (rank 16)
L9.attn	9.688	16.281	0.714 (rank 16)
L9.ffn	9.844	16.523	0.715 (rank 16)
L10.attn	9.832	16.555	0.715 (rank 16)
L10.ffn	10.124	16.779	0.717 (rank 16)
L11.attn	10.139	16.805	0.714 (rank 16)
L11.ffn	10.741	16.916	0.713 (rank 16)

35M_seeds (SimpleStories/SimpleStories-35M)

location	TwoNN	MLE	val recon m=16
embed	23.000	28.573	0.866 (rank 16)
L0.attn	23.855	28.275	0.889 (rank 16)
L0.ffn	15.243	27.655	0.804 (rank 16)
L1.attn	15.390	27.768	0.806 (rank 16)
L1.ffn	12.394	24.061	0.782 (rank 16)
L2.attn	12.448	23.696	0.781 (rank 16)
L2.ffn	11.188	20.273	0.767 (rank 16)
L3.attn	10.997	20.091	0.767 (rank 16)
L3.ffn	10.393	19.375	0.755 (rank 16)
L4.attn	10.332	19.209	0.752 (rank 16)
L4.ffn	10.035	18.590	0.739 (rank 16)
L5.attn	9.900	18.511	0.740 (rank 16)
L5.ffn	9.898	17.878	0.734 (rank 16)
L6.attn	9.870	17.718	0.729 (rank 16)
L6.ffn	10.133	17.067	0.726 (rank 16)

location	TwoNN	MLE	val recon m=16
L7.attn	9.956	16.577	0.718 (rank 16)
L7.ffn	9.941	16.389	0.712 (rank 16)
L8.attn	9.989	16.329	0.713 (rank 16)
L8.ffn	9.950	16.504	0.715 (rank 16)
L9.attn	9.688	16.281	0.711 (rank 16)
L9.ffn	9.844	16.523	0.714 (rank 16)
L10.attn	9.832	16.555	0.713 (rank 16)
L10.ffn	10.124	16.779	0.717 (rank 16)
L11.attn	10.139	16.805	0.715 (rank 16)
L11.ffn	10.741	16.916	0.712 (rank 16)

5M (SimpleStories/SimpleStories-5M)

location	TwoNN	MLE	val recon m=8	val recon m=16	val recon m=32
embed	18.367	21.788	0.832 (rank 8)	0.786 (rank 16)	0.743 (rank 32)
L0.attn	18.159	21.976	0.852 (rank 8)	0.812 (rank 16)	0.765 (rank 32)
L0.ffn	14.250	18.821	0.783 (rank 8)	0.750 (rank 16)	0.703 (rank 32)
L1.attn	14.342	18.790	0.783 (rank 8)	0.747 (rank 16)	0.703 (rank 32)
L1.ffn	14.095	17.632	0.781 (rank 8)	0.738 (rank 16)	0.690 (rank 32)
L2.attn	14.010	17.550	0.779 (rank 8)	0.737 (rank 16)	0.685 (rank 32)
L2.ffn	13.163	16.629	0.771 (rank 8)	0.725 (rank 16)	0.664 (rank 32)
L3.attn	13.147	16.373	0.773 (rank 8)	0.722 (rank 16)	0.660 (rank 32)
L3.ffn	12.594	16.125	0.768 (rank 8)	0.714 (rank 16)	0.646 (rank 32)
L4.attn	12.470	16.183	0.765 (rank 8)	0.711 (rank 16)	0.636 (rank 32)
L4.ffn	12.731	16.964	0.776 (rank 8)	0.715 (rank 16)	0.637 (rank 32)
L5.attn	12.667	16.941	0.767 (rank 8)	0.705 (rank 16)	0.620 (rank 32)
L5.ffn	14.109	17.737	0.769 (rank 8)	0.705 (rank 16)	0.607 (rank 32)

5M_prefix2 (SimpleStories/SimpleStories-5M)

location	TwoNN	MLE	val recon m=16
embed	18.367	21.788	0.785 (rank 16)
L0.attn	18.433	21.762	0.798 (rank 16)
L0.ffn	14.628	18.799	0.750 (rank 16)
L1.attn	14.715	18.835	0.744 (rank 16)
L1.ffn	14.619	17.816	0.732 (rank 16)
L2.attn	14.605	17.825	0.735 (rank 16)
L2.ffn	13.523	16.819	0.723 (rank 16)
L3.attn	13.298	16.477	0.717 (rank 16)
L3.ffn	12.991	16.204	0.707 (rank 16)
L4.attn	12.902	16.273	0.706 (rank 16)
L4.ffn	13.298	16.698	0.715 (rank 16)
L5.attn	13.386	16.850	0.709 (rank 16)
L5.ffn	14.251	17.093	0.705 (rank 16)

TinyLlama (TinyLlama/TinyLlama-1.1B-Chat-v1.0)

location	TwoNN	MLE	val recon m=16
embed	234	32.065	6.084 (rank 16)
L0.attn	94.784	37.765	4.286 (rank 16)
L0.ffn	18.079	31.489	0.976 (rank 16)
L3.ffn	11.911	26.038	0.789 (rank 16)
L7.ffn	10.327	22.550	0.764 (rank 16)
L11.ffn	12.867	23.396	0.770 (rank 16)
L15.ffn	8.481	16.980	0.774 (rank 16)
L21.ffn	9.465	18.466	0.761 (rank 16)

TinyLlama_all (TinyLlama/TinyLlama-1.1B-Chat-v1.0)

location	TwoNN	MLE	val recon m=16
embed	234	32.065	6.084 (rank 16)

location	TwoNN	MLE	val recon m=16
L0.attn	94.784	37.765	4.282 (rank 16)
L0.ffn	18.079	31.489	0.974 (rank 16)
L1.attn	16.461	30.534	0.915 (rank 16)
L1.ffn	14.716	27.207	0.838 (rank 16)
L2.attn	15.036	27.849	0.825 (rank 16)
L2.ffn	15.244	30.256	0.817 (rank 16)
L3.attn	15.355	30.845	0.805 (rank 16)
L3.ffn	11.911	26.038	0.790 (rank 16)
L4.attn	12.464	27.062	0.786 (rank 16)
L4.ffn	11.520	26.684	0.774 (rank 16)
L5.attn	11.702	26.673	0.772 (rank 16)
L5.ffn	11.076	25.488	0.769 (rank 16)
L6.attn	11.231	26.003	0.766 (rank 16)
L6.ffn	10.726	24.191	0.770 (rank 16)
L7.attn	10.723	24.274	0.770 (rank 16)
L7.ffn	10.327	22.550	0.764 (rank 16)
L8.attn	10.442	22.710	0.763 (rank 16)
L8.ffn	10.962	23.905	0.768 (rank 16)
L9.attn	11.348	24.256	0.766 (rank 16)
L9.ffn	11.070	23.240	0.768 (rank 16)
L10.attn	11.267	23.263	0.766 (rank 16)
L10.ffn	12.254	23.563	0.767 (rank 16)
L11.attn	12.445	23.679	0.767 (rank 16)
L11.ffn	12.867	23.396	0.770 (rank 16)
L12.attn	12.944	23.486	0.769 (rank 16)
L12.ffn	11.704	21.333	0.772 (rank 16)
L13.attn	11.873	21.441	0.771 (rank 16)
L13.ffn	11.177	19.719	0.764 (rank 16)
L14.attn	11.114	19.563	0.760 (rank 16)
L14.ffn	10.010	18.826	0.761 (rank 16)
L15.attn	9.900	18.706	0.762 (rank 16)
L15.ffn	8.481	16.980	0.774 (rank 16)
L16.attn	8.591	17.081	0.774 (rank 16)
L16.ffn	7.939	15.171	0.787 (rank 16)
L17.attn	7.952	15.215	0.785 (rank 16)
L17.ffn	7.916	15.268	0.786 (rank 16)
L18.attn	7.962	15.259	0.784 (rank 16)
L18.ffn	8.006	15.521	0.789 (rank 16)
L19.attn	8.000	15.543	0.787 (rank 16)
L19.ffn	8.126	16.240	0.789 (rank 16)
L20.attn	8.118	16.247	0.784 (rank 16)
L20.ffn	8.395	17.297	0.785 (rank 16)
L21.attn	8.449	17.337	0.781 (rank 16)
L21.ffn	9.465	18.466	0.761 (rank 16)

TinyLlama_m64 (TinyLlama/TinyLlama-1.1B-Chat-v1.0)

location	TwoNN	MLE	val recon m=64
embed	234	32.065	5.959 (rank 64)
L0.ffn	18.079	31.489	0.934 (rank 64)
L3.ffn	11.911	26.038	0.730 (rank 64)
L11.ffn	12.867	23.396	0.658 (rank 64)
L21.ffn	9.465	18.466	0.681 (rank 64)

2. Samples versus α (mean over locations; bands in the last rows)

35M, m = 16

α / band	nearest real	recon	tangent resid	u (mean)	u (min over loc)	coverage	ESS	ball multiplicity
0.100	0.617	0.127	0.567	1.114	1.004	0.992	2724	1.088
0.200	0.621	0.130	0.571	1.120	1.009	0.989	2697	1.200
0.300	0.622	0.136	0.568	1.117	1.095	0.985	2638	1.278
0.500	0.662	0.169	0.596	1.093	1.069	0.980	2479	1.337
0.700	0.731	0.222	0.649	1.052	1.021	0.973	2283	1.322
1.000	0.868	0.344	0.761	0.958	0.922	0.929	1926	1.421
1.500	1.181	0.698	1.024	0.721	0.680	0.119	1288	1.866
held-out real	0.896	0.752	0.808	0.997	0.990	-	-	-
real + 0.5x noise	1.022	0.909	0.945	0.905	0.901	-	-	-
real + 1x noise	1.330	1.270	1.265	0.674	0.666	-	-	-

35M_seeds, m = 16

α / band	nearest real	recon	tangent resid	u (mean)	u (min over loc)	coverage	ESS	ball multiplicity
0.100	0.616	0.128	0.566	1.114	1.006	0.992	2817	1.073
0.200	0.622	0.130	0.571	1.118	1.015	0.989	2790	1.172
0.300	0.621	0.136	0.567	1.118	1.096	0.985	2732	1.234
0.500	0.660	0.168	0.595	1.093	1.071	0.980	2564	1.282
0.700	0.729	0.221	0.648	1.053	1.027	0.973	2338	1.310
1.000	0.865	0.339	0.758	0.961	0.923	0.932	1966	1.384
1.500	1.175	0.685	1.019	0.726	0.683	0.122	1324	1.881
held-out real	0.896	0.751	0.808	0.997	0.990	-	-	-
real + 0.5x noise	1.022	0.908	0.945	0.905	0.901	-	-	-
real + 1x noise	1.330	1.268	1.265	0.674	0.666	-	-	-

5M, m = 8

α / band	nearest real	recon	tangent resid	u (mean)	u (min over loc)	coverage	ESS	ball multiplicity
0.100	0.656	0.092	0.605	1.160	1.139	0.995	5757	1.017
0.200	0.657	0.093	0.606	1.159	1.136	0.994	5661	1.034
0.300	0.664	0.097	0.612	1.158	1.136	0.992	5536	1.047
0.500	0.683	0.108	0.625	1.151	1.129	0.988	5103	1.113
0.700	0.710	0.127	0.646	1.135	1.112	0.983	4222	1.244
1.000	0.770	0.184	0.697	1.081	1.049	0.976	2575	1.695
1.500	0.949	0.354	0.853	0.919	0.866	0.950	1106	3.039
held-out real	0.914	0.784	0.863	0.998	0.994	-	-	-
real + 0.5x noise	1.033	0.934	0.987	0.913	0.910	-	-	-
real + 1x noise	1.320	1.278	1.280	0.694	0.685	-	-	-

5M_prefix2, m = 16

α / band	nearest real	recon	tangent resid	u (mean)	u (min over loc)	coverage	ESS	ball multiplicity
0.100	0.590	0.109	0.545	1.119	1.099	0.992	3369	1.028
0.200	0.598	0.114	0.550	1.116	1.096	0.989	3361	1.052
0.300	0.613	0.123	0.559	1.109	1.089	0.986	3335	1.065
0.500	0.664	0.154	0.592	1.086	1.067	0.976	3212	1.094
0.700	0.735	0.204	0.643	1.049	1.029	0.967	3013	1.124
1.000	0.876	0.320	0.749	0.962	0.938	0.932	2462	1.258
1.500	1.198	0.667	1.010	0.730	0.683	0.111	1374	1.914
held-out real	0.920	0.733	0.823	0.998	0.993	-	-	-
real + 0.5x noise	1.038	0.888	0.951	0.913	0.910	-	-	-
real + 1x noise	1.330	1.241	1.251	0.695	0.686	-	-	-

TinyLlama, m = 16

α / band	nearest real	recon	tangent resid	u (mean)	u (min over loc)	coverage	ESS	ball multiplicity
0.100	1.135	0.167	1.042	1.225	1.135	0.985	789	1.000
0.200	1.143	0.173	1.050	1.223	1.131	0.983	796	1.000
0.300	1.162	0.184	1.068	1.218	1.125	0.983	806	1.001
0.500	1.214	0.223	1.121	1.206	1.110	0.977	849	1.016
0.700	1.281	0.277	1.187	1.190	1.085	0.969	888	1.066

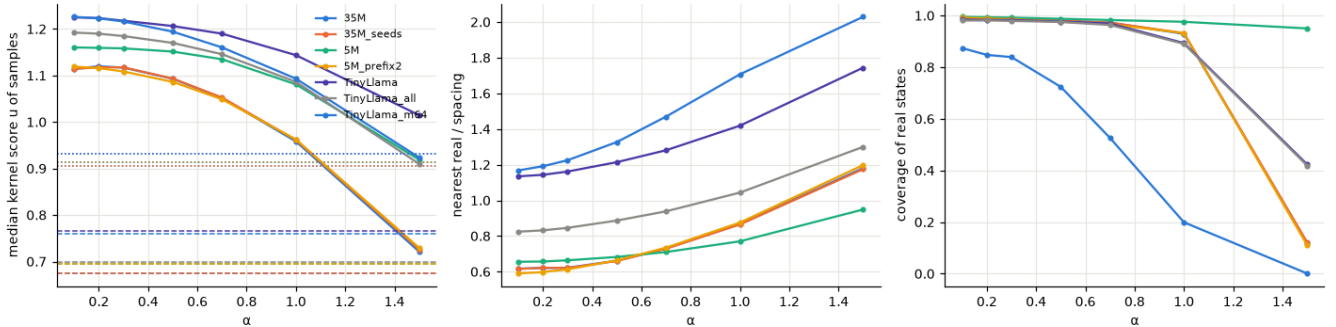
α / band	nearest real	recon	tangent resid	u (mean)	u (min over loc)	coverage	ESS	ball multiplicity
1.000	1.419	0.398	1.318	1.144	1.005	0.894	835	1.281
1.500	1.744	0.739	1.633	1.015	0.764	0.424	497	2.075
held-out real	2.035	1.901	1.970	0.998	0.996	-	-	-
real + 0.5x noise	2.141	2.021	2.084	0.932	0.907	-	-	-
real + 1x noise	2.403	2.300	2.355	0.767	0.675	-	-	-

TinyLlama_all, m = 16

α / band	nearest real	recon	tangent resid	u (mean)	u (min over loc)	coverage	ESS	ball multiplicity
0.100	0.824	0.091	0.715	1.192	1.139	0.981	762	1.000
0.200	0.832	0.096	0.723	1.190	1.134	0.981	778	1.000
0.300	0.846	0.104	0.738	1.185	1.131	0.979	807	1.002
0.500	0.887	0.128	0.780	1.170	1.112	0.975	901	1.018
0.700	0.939	0.159	0.833	1.146	1.085	0.964	1025	1.077
1.000	1.044	0.225	0.934	1.086	1.024	0.891	1085	1.315
1.500	1.300	0.410	1.176	0.910	0.778	0.419	654	2.260
held-out real	1.077	0.981	1.002	0.999	0.996	-	-	-
real + 0.5x noise	1.205	1.124	1.140	0.914	0.904	-	-	-
real + 1x noise	1.514	1.456	1.463	0.700	0.666	-	-	-

TinyLlama_m64, m = 64

α / band	nearest real	recon	tangent resid	u (mean)	u (min over loc)	coverage	ESS	ball multiplicity
0.100	1.168	0.085	1.046	1.226	1.114	0.874	1718	1.000
0.200	1.191	0.089	1.069	1.223	1.108	0.848	1667	1.000
0.300	1.225	0.095	1.096	1.216	1.094	0.839	1596	1.000
0.500	1.327	0.115	1.185	1.194	1.059	0.724	1419	1.000
0.700	1.470	0.145	1.308	1.160	1.011	0.526	1234	1.001
1.000	1.707	0.206	1.513	1.093	0.916	0.199	991	1.002
1.500	2.029	0.348	1.790	0.924	0.667	0.001	646	1.057
held-out real	2.006	1.792	1.872	0.998	0.996	-	-	-
real + 0.5x noise	2.113	1.921	1.992	0.931	0.909	-	-	-
real + 1x noise	2.383	2.219	2.277	0.760	0.678	-	-	-



Dotted / dashed lines: kernel score of real states displaced by 0.5x / 1x spacing (the off-support bands).

3. Latent dimension and seed sensitivity (35M)

m	α	u mean	u range (loc × seed)	nearest real	coverage	ESS	val recon range
16	0.300	1.117	1.095–1.205	0.622	0.985	2638	0.711–0.892
16	0.500	1.093	1.069–1.191	0.662	0.980	2479	0.711–0.892
16	1.000	0.958	0.922–1.080	0.868	0.929	1926	0.711–0.892

3. Latent dimension and seed sensitivity (35M_seeds)

m	α	u mean	u range (loc × seed)	nearest real	coverage	ESS	val recon range
16	0.300	1.118	1.096–1.189	0.621	0.985	2732	0.709–0.895
16	0.500	1.093	1.071–1.183	0.660	0.980	2564	0.709–0.895
16	1.000	0.961	0.923–1.081	0.865	0.932	1966	0.709–0.895

3. Latent dimension and seed sensitivity (5M)

m	α	u mean	u range (loc $\times$ seed)	nearest real	coverage	ESS	val recon range
8	0.300	1.158	1.136-1.217	0.664	0.992	5536	0.763-0.856
8	0.500	1.151	1.129-1.209	0.683	0.988	5103	0.763-0.856
8	1.000	1.081	1.049-1.152	0.770	0.976	2575	0.763-0.856
16	0.300	1.110	1.094-1.172	0.614	0.988	3436	0.704-0.816
16	0.500	1.087	1.071-1.152	0.662	0.979	3336	0.704-0.816
16	1.000	0.965	0.944-1.025	0.867	0.944	2574	0.704-0.816
32	0.300	1.058	1.039-1.110	0.535	0.958	1490	0.607-0.768
32	0.500	1.010	0.993-1.061	0.628	0.927	1138	0.607-0.768
32	1.000	0.786	0.760-0.827	1.001	0.241	569	0.607-0.768

3. Latent dimension and seed sensitivity (5M_prefix2)

m	α	u mean	u range (loc $\times$ seed)	nearest real	coverage	ESS	val recon range
16	0.300	1.109	1.089-1.168	0.613	0.986	3335	0.703-0.800
16	0.500	1.086	1.067-1.138	0.664	0.976	3212	0.703-0.800
16	1.000	0.962	0.938-1.015	0.876	0.932	2462	0.703-0.800

3. Latent dimension and seed sensitivity (TinyLlama)

m	α	u mean	u range (loc $\times$ seed)	nearest real	coverage	ESS	val recon range
16	0.300	1.218	1.125-1.375	1.162	0.983	806	0.761-6.084
16	0.500	1.206	1.110-1.388	1.214	0.977	849	0.761-6.084
16	1.000	1.144	1.005-1.418	1.419	0.894	835	0.761-6.084

4. Sampler variants and loss ablation (5M, $\alpha = 0.3$)

Sampler variants (same decoder): support radius rule (global = $\alpha \times \text{median } \rho$ for all anchors, local = $\alpha \times \rho_i$), anchor probability $\propto R^p$ and whether the volume reweighting is applied. anchors = participation ratio of the anchor distribution.

location	variant	anchors	ESS	coverage	nearest real	recon	u
embed	tangent charts (cross-check)	-	2000	0.738	0.274	0.805	0.966
embed	radius=global, p=m, reweight=True	3546	4898	0.991	0.710	0.107	1.215
embed	radius=global, p=0, reweight=False	3546	16000	0.992	0.775	0.069	1.241
embed	radius=local, p=m, reweight=True	336	8555	0.960	0.668	0.134	1.193
embed	radius=local, p=0, reweight=True	3546	432	0.958	0.666	0.132	1.183
embed	radius=local, p=0, reweight=False	3546	16000	0.997	0.774	0.068	1.240
L2.ffn	tangent charts (cross-check)	-	2000	0.866	0.274	0.770	0.971
L2.ffn	radius=global, p=m, reweight=True	3546	6063	0.991	0.663	0.093	1.148
L2.ffn	radius=global, p=0, reweight=False	3546	16000	0.992	0.719	0.064	1.218
L2.ffn	radius=local, p=m, reweight=True	755	7868	0.970	0.626	0.115	1.126
L2.ffn	radius=local, p=0, reweight=True	3546	1033	0.970	0.626	0.113	1.127
L2.ffn	radius=local, p=0, reweight=False	3546	16000	0.995	0.720	0.064	1.218
L5.ffn	tangent charts (cross-check)	-	2000	0.836	0.274	0.756	0.975
L5.ffn	radius=global, p=m, reweight=True	3546	6780	0.995	0.694	0.073	1.165
L5.ffn	radius=global, p=0, reweight=False	3546	16000	0.995	0.725	0.052	1.201
L5.ffn	radius=local, p=m, reweight=True	104	4807	0.964	0.384	0.126	1.055
L5.ffn	radius=local, p=0, reweight=True	3546	40.125	0.973	0.437	0.153	1.090
L5.ffn	radius=local, p=0, reweight=False	3546	16000	0.996	0.724	0.052	1.201

Loss / architecture variants (each refitted, then sampled at $\alpha = 0.3$).

location	variant	val recon	train recon	ESS	coverage	nearest real	recon	u
embed	default	0.833	0.670	4898	0.991	0.710	0.107	1.215
embed	no_geom	0.843	0.578	4374	0.983	0.612	0.133	1.150
embed	no_curv	0.832	0.670	4868	0.990	0.704	0.106	1.216
embed	strong_geom	0.844	0.707	5880	0.992	0.780	0.078	1.237
embed	wide	0.827	0.655	4027	0.990	0.690	0.108	1.209

location	variant	val recon	train recon	ESS	coverage	nearest real	recon	u
embed	epochs_900	0.894	0.511	1967	0.960	0.575	0.345	1.112
L2.ffn	default	0.772	0.582	6063	0.991	0.663	0.093	1.148
L2.ffn	no_geom	0.781	0.501	4899	0.991	0.578	0.137	1.104
L2.ffn	no_curv	0.771	0.582	6083	0.992	0.662	0.090	1.149
L2.ffn	strong_geom	0.779	0.612	5422	0.992	0.682	0.075	1.160
L2.ffn	wide	0.772	0.565	6077	0.993	0.640	0.095	1.141
L2.ffn	epochs_900	0.821	0.435	2320	0.972	0.558	0.350	1.067
L5.ffn	default	0.767	0.523	6780	0.995	0.694	0.073	1.165
L5.ffn	no_geom	0.778	0.459	7407	0.997	0.630	0.101	1.126
L5.ffn	no_curv	0.768	0.523	6802	0.993	0.690	0.072	1.166
L5.ffn	strong_geom	0.778	0.544	6167	0.992	0.705	0.063	1.180
L5.ffn	wide	0.771	0.507	6003	0.995	0.675	0.080	1.157
L5.ffn	epochs_900	0.813	0.397	2078	0.988	0.551	0.273	1.083

5. Propagation through the real model: images vs independent destination fits

Samples at the embedding are pushed through the real Transformer stretch embed $\rightarrow$ dst (fixed prefix context) and scored by the destination manifold fitted only on real destination states. Controls: images of held-out real states, and of real states displaced by 0.5 $\times$ / 1 $\times$ spacing *before* propagation.

35M: embed $\rightarrow$ L0.attn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.833	0.425	0.793	1.173
sample $\alpha=0.5$	0.729	0.420	0.690	1.173
sample $\alpha=1.0$	0.880	0.649	0.822	1.056
real held-out images	1.028	0.892	0.981	0.990
real + 0.5x noise images	1.140	1.046	1.103	0.905
real + 1x noise images	1.484	1.427	1.436	0.703
band: held-out real	1.028	0.892	0.981	0.990
band: real + 0.5x noise	1.144	1.041	1.104	0.909
band: real + 1x noise	1.452	1.409	1.410	0.716

35M: embed $\rightarrow$ L0.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.843	0.814	0.759	1.009
sample $\alpha=0.5$	0.756	0.662	0.682	1.046
sample $\alpha=1.0$	0.820	0.653	0.749	1.070
real held-out images	0.937	0.802	0.878	0.996
real + 0.5x noise images	0.968	0.829	0.909	1.014
real + 1x noise images	2.279	1.901	2.198	0.994
band: held-out real	0.937	0.802	0.878	0.996
band: real + 0.5x noise	1.055	0.968	1.004	0.917
band: real + 1x noise	1.362	1.307	1.320	0.677

35M: embed $\rightarrow$ L5.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.837	0.656	0.741	1.012
sample $\alpha=0.5$	0.745	0.594	0.646	1.039
sample $\alpha=1.0$	0.802	0.579	0.699	1.080
real held-out images	0.867	0.734	0.784	0.996
real + 0.5x noise images	0.877	0.707	0.789	1.028
real + 1x noise images	1.006	0.734	0.967	1.060
band: held-out real	0.867	0.734	0.784	0.996
band: real + 0.5x noise	1.001	0.894	0.926	0.904
band: real + 1x noise	1.320	1.257	1.244	0.670

35M: embed $\rightarrow$ L11.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.882	0.670	0.752	1.003
sample $\alpha=0.5$	0.805	0.629	0.682	1.009
sample $\alpha=1.0$	0.863	0.628	0.745	1.018
real held-out images	0.900	0.714	0.790	0.998
real + 0.5x noise images	0.917	0.702	0.811	1.005
real + 1x noise images	0.988	0.728	0.918	1.017
band: held-out real	0.900	0.714	0.790	0.998
band: real + 0.5x noise	1.022	0.872	0.924	0.904
band: real + 1x noise	1.332	1.237	1.249	0.669

5M: embed $\rightarrow$ L0.attn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.724	0.231	0.667	1.202
sample $\alpha=0.5$	0.738	0.253	0.682	1.196
sample $\alpha=1.0$	0.791	0.351	0.726	1.129
real held-out images	0.965	0.856	0.904	0.994
real + 0.5x noise images	1.090	1.003	1.033	0.911
real + 1x noise images	1.396	1.361	1.358	0.709
band: held-out real	0.965	0.856	0.904	0.994
band: real + 0.5x noise	1.087	1.003	1.030	0.914
band: real + 1x noise	1.379	1.341	1.338	0.721

5M: embed $\rightarrow$ L0.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.835	0.734	0.772	1.033
sample $\alpha=0.5$	0.841	0.722	0.777	1.038
sample $\alpha=1.0$	0.855	0.718	0.781	1.030
real held-out images	0.902	0.783	0.856	1.001
real + 0.5x noise images	1.005	0.904	0.962	0.945
real + 1x noise images	1.246	1.170	1.213	0.783
band: held-out real	0.902	0.783	0.856	1.001
band: real + 0.5x noise	1.026	0.940	0.986	0.914
band: real + 1x noise	1.320	1.281	1.283	0.695

5M: embed $\rightarrow$ L2.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.900	0.751	0.824	0.996
sample $\alpha=0.5$	0.896	0.746	0.822	0.997
sample $\alpha=1.0$	0.899	0.750	0.820	0.994
real held-out images	0.895	0.772	0.851	0.998
real + 0.5x noise images	0.986	0.845	0.942	0.982
real + 1x noise images	1.172	1.039	1.130	0.896
band: held-out real	0.895	0.772	0.851	0.998
band: real + 0.5x noise	1.018	0.924	0.972	0.912
band: real + 1x noise	1.309	1.266	1.268	0.691

5M: embed $\rightarrow$ L5.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.912	0.761	0.837	0.991
sample $\alpha=0.5$	0.915	0.763	0.842	0.992
sample $\alpha=1.0$	0.922	0.769	0.848	0.993
real held-out images	0.920	0.767	0.858	0.998
real + 0.5x noise images	0.982	0.819	0.935	0.999
real + 1x noise images	1.057	0.875	1.017	0.995
band: held-out real	0.920	0.767	0.858	0.998
band: real + 0.5x noise	1.031	0.916	0.981	0.910
band: real + 1x noise	1.325	1.260	1.273	0.697

5M_prefix2: embed → L0.attn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.651	0.354	0.607	1.141
sample $\alpha=0.5$	0.691	0.414	0.634	1.113
sample $\alpha=1.0$	0.866	0.654	0.774	0.992
real held-out images	0.953	0.800	0.871	0.993
real + 0.5x noise images	1.083	0.949	1.004	0.914
real + 1x noise images	1.361	1.302	1.294	0.719
band: held-out real	0.953	0.800	0.871	0.993
band: real + 0.5x noise	1.081	0.953	0.998	0.914
band: real + 1x noise	1.366	1.301	1.296	0.712

5M_prefix2: embed → L0.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.737	0.598	0.669	1.037
sample $\alpha=0.5$	0.762	0.623	0.688	1.026
sample $\alpha=1.0$	0.900	0.761	0.805	0.960
real held-out images	0.907	0.751	0.828	1.000
real + 0.5x noise images	1.009	0.867	0.940	0.947
real + 1x noise images	1.260	1.147	1.190	0.790
band: held-out real	0.907	0.751	0.828	1.000
band: real + 0.5x noise	1.030	0.906	0.948	0.914
band: real + 1x noise	1.322	1.256	1.258	0.696

5M_prefix2: embed → L2.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.807	0.668	0.711	1.000
sample $\alpha=0.5$	0.826	0.676	0.726	0.997
sample $\alpha=1.0$	0.921	0.744	0.818	0.977
real held-out images	0.917	0.724	0.815	0.998
real + 0.5x noise images	0.990	0.803	0.908	0.979
real + 1x noise images	1.165	0.999	1.101	0.897
band: held-out real	0.917	0.724	0.815	0.998
band: real + 0.5x noise	1.029	0.876	0.944	0.914
band: real + 1x noise	1.321	1.224	1.241	0.695

5M_prefix2: embed → L5.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.857	0.662	0.715	0.995
sample $\alpha=0.5$	0.873	0.671	0.744	0.995
sample $\alpha=1.0$	0.942	0.729	0.818	0.988
real held-out images	0.928	0.708	0.815	0.996
real + 0.5x noise images	0.984	0.746	0.878	0.999
real + 1x noise images	1.064	0.826	0.974	0.993
band: held-out real	0.928	0.708	0.815	0.996
band: real + 0.5x noise	1.042	0.861	0.942	0.912
band: real + 1x noise	1.339	1.225	1.246	0.693

TinyLlama: embed → L0.attn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	3.115	2.530	3.032	1.158
sample $\alpha=0.5$	3.129	2.544	3.054	1.160
sample $\alpha=1.0$	3.231	2.547	3.157	1.191
real held-out images	4.556	4.282	4.502	0.996
real + 0.5x noise images	4.597	4.324	4.537	0.986
real + 1x noise images	4.658	4.395	4.605	0.978
band: held-out real	4.556	4.282	4.502	0.996

set	nearest real	recon	tangent resid	u
band: real + 0.5x noise	4.601	4.332	4.547	0.981
band: real + 1x noise	4.701	4.442	4.644	0.967

TinyLlama: embed → L0.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	1.807	1.465	1.706	1.011
sample $\alpha=0.5$	1.778	1.417	1.670	1.018
sample $\alpha=1.0$	1.828	1.483	1.715	1.038
real held-out images	1.053	0.974	1.016	0.998
real + 0.5x noise images	1.062	0.995	1.025	0.994
real + 1x noise images	1.160	1.133	1.121	0.989
band: held-out real	1.053	0.974	1.016	0.998
band: real + 0.5x noise	1.170	1.107	1.138	0.940
band: real + 1x noise	1.484	1.431	1.446	0.773

TinyLlama: embed → L7.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.974	0.735	0.885	1.008
sample $\alpha=0.5$	0.968	0.735	0.877	1.007
sample $\alpha=1.0$	0.980	0.741	0.883	1.002
real held-out images	0.858	0.764	0.798	0.999
real + 0.5x noise images	0.863	0.763	0.803	1.000
real + 1x noise images	0.879	0.764	0.819	1.002
band: held-out real	0.858	0.764	0.798	0.999
band: real + 0.5x noise	0.992	0.914	0.938	0.907
band: real + 1x noise	1.313	1.262	1.272	0.675

TinyLlama: embed → L21.ffn

set	nearest real	recon	tangent resid	u
sample $\alpha=0.25$	0.877	0.688	0.737	1.012
sample $\alpha=0.5$	0.879	0.686	0.744	1.014
sample $\alpha=1.0$	0.887	0.678	0.752	1.014
real held-out images	0.864	0.761	0.755	0.998
real + 0.5x noise images	0.866	0.762	0.759	0.998
real + 1x noise images	0.873	0.761	0.770	0.999
band: held-out real	0.864	0.761	0.755	0.998
band: real + 0.5x noise	0.997	0.913	0.904	0.910
band: real + 1x noise	1.318	1.261	1.247	0.689

6. Real-occurrence clouds (random token occurrences in real stories, split by story)

Same fit and sampling protocol on residual states of real token occurrences (20k occurrences, 8 per story, ≤ 256 tokens per story); duplicates and isolated outliers dropped from the training split. Propagation uses the tokenwise FFN maps $L_k.\text{attn} \rightarrow L_k.\text{ffn}$.

35M_stories (SimpleStories/SimpleStories-35M: 20000 occurrences from 2500 stories, 2000 held out)

location	train states	TwoNN	MLE	val recon
L0.attn	17169	8.641	2.014	1.039
L0.ffn	17134	8.388	1.946	1.267
L1.attn	17933	5.659	4.501	0.942
L1.ffn	17933	5.545	4.348	0.991
L2.attn	17933	7.024	6.105	0.890
L2.ffn	17933	7.030	5.949	0.915
L3.attn	17933	8.436	7.844	0.853
L3.ffn	17933	8.377	7.453	0.865
L4.attn	17933	9.793	8.837	0.839

location	train states	TwoNN	MLE	val recon
L4.ffn	17933	9.604	8.384	0.840
L5.attn	17933	10.431	9.824	0.821
L5.ffn	17933	10.267	9.454	0.826
L6.attn	17933	10.588	10.323	0.804
L6.ffn	17933	10.330	10.024	0.805
L7.attn	17933	11.219	11.512	0.792
L7.ffn	17933	11.080	11.327	0.792
L8.attn	17933	11.423	12.275	0.782
L8.ffn	17933	11.279	12.195	0.783
L9.attn	17933	11.951	13.551	0.781
L9.ffn	17933	11.991	13.623	0.779
L10.attn	17933	12.795	15.012	0.772
L10.ffn	17933	12.912	15.344	0.775
L11.attn	17933	13.301	16.178	0.769
L11.ffn	17933	13.743	17.434	0.764

α / band	nearest real	recon	u (mean)	u (min over loc)	coverage	ESS
0.100	0.888	0.238	1.120	1.012	0.784	2484
0.300	0.908	0.251	1.115	1.012	0.777	2477
0.500	0.949	0.272	1.107	1.012	0.757	2418
0.700	1.006	0.305	1.094	1.010	0.720	2254
1.000	1.104	0.361	1.059	1.007	0.578	1754
1.500	1.311	0.496	0.967	0.875	0.214	996
held-out real	0.809	0.854	0.998	0.996	-	-
real + 0.5x noise	0.949	0.998	0.942	0.912	-	-
real + 1x noise	1.277	1.342	0.794	0.703	-	-

L0.attn → L0.ffn

set	nearest real	recon	u
sample $\alpha=0.3$	1.575	2.821	1.003
sample $\alpha=0.7$	1.888	3.024	1.004
sample $\alpha=1.0$	2.036	3.098	1.003
real held-out images	0.829	1.267	1.000
real + 1x noise images	1.258	1.767	0.999
band: held-out real	0.829	1.267	1.000
band: real + 0.5x noise	0.967	1.392	0.996
band: real + 1x noise	1.280	1.719	0.989

L1.attn → L1.ffn

set	nearest real	recon	u
sample $\alpha=0.3$	0.942	0.854	1.056
sample $\alpha=0.7$	1.004	0.920	1.063
sample $\alpha=1.0$	1.075	1.013	1.058
real held-out images	0.699	0.991	0.999
real + 1x noise images	1.138	1.362	0.960
band: held-out real	0.699	0.991	0.999
band: real + 0.5x noise	0.858	1.124	0.978
band: real + 1x noise	1.209	1.453	0.916

L2.attn → L2.ffn

set	nearest real	recon	u
sample $\alpha=0.3$	0.910	0.701	1.064
sample $\alpha=0.7$	0.972	0.800	1.053
sample $\alpha=1.0$	1.066	0.915	1.039
real held-out images	0.739	0.915	0.999
real + 1x noise images	1.165	1.286	0.927
band: held-out real	0.739	0.915	0.999
band: real + 0.5x noise	0.891	1.052	0.964

set	nearest real	recon	u
band: real + 1x noise	1.238	1.382	0.863

L3.attn → L3.ffn

set	nearest real	recon	u
sample $\alpha=0.3$	0.832	0.598	1.058
sample $\alpha=0.7$	0.919	0.718	1.047
sample $\alpha=1.0$	1.030	0.844	1.030
real held-out images	0.777	0.865	0.998
real + 1x noise images	1.173	1.231	0.893
band: held-out real	0.777	0.865	0.998
band: real + 0.5x noise	0.923	1.004	0.951
band: real + 1x noise	1.258	1.344	0.818

5M_stories (SimpleStories/SimpleStories-5M: 20000 occurrences from 2500 stories, 2000 held out)

location	train states	TwoNN	MLE	val recon
L0.attn	16877	8.327	2.005	1.167
L0.ffn	16788	8.199	2.081	1.225
L1.attn	17933	5.294	3.782	0.899
L1.ffn	17933	5.291	3.999	0.914
L2.attn	17933	6.312	5.015	0.858
L2.ffn	17933	6.302	5.195	0.852
L3.attn	17933	7.109	6.569	0.809
L3.ffn	17933	7.152	6.813	0.800
L4.attn	17933	9.227	9.133	0.773
L4.ffn	17933	9.333	9.464	0.771
L5.attn	17933	11.614	12.317	0.760
L5.ffn	17933	12.209	13.917	0.763

α / band	nearest real	recon	u (mean)	u (min over loc)	coverage	ESS
0.100	0.897	0.293	1.081	1.016	0.676	3014
0.300	0.918	0.302	1.079	1.014	0.652	2991
0.500	0.955	0.316	1.073	1.015	0.608	2830
0.700	1.006	0.340	1.065	1.015	0.546	2556
1.000	1.088	0.377	1.043	1.015	0.425	2023
1.500	1.261	0.465	0.978	0.887	0.204	1281
held-out real	0.760	0.883	0.998	0.997	-	-
real + 0.5x noise	0.905	1.031	0.963	0.929	-	-
real + 1x noise	1.239	1.387	0.862	0.749	-	-

L0.attn → L0.ffn

set	nearest real	recon	u
sample $\alpha=0.3$	1.542	2.166	1.009
sample $\alpha=0.7$	1.672	2.317	1.010
sample $\alpha=1.0$	1.719	2.376	1.010
real held-out images	0.788	1.225	0.999
real + 1x noise images	1.245	1.709	0.990
band: held-out real	0.787	1.225	0.999
band: real + 0.5x noise	0.927	1.368	0.995
band: real + 1x noise	1.254	1.747	0.988

L1.attn → L1.ffn

set	nearest real	recon	u
sample $\alpha=0.3$	0.885	0.811	1.033
sample $\alpha=0.7$	0.944	0.879	1.029
sample $\alpha=1.0$	0.994	0.942	1.021
real held-out images	0.665	0.914	0.999
real + 1x noise images	1.153	1.340	0.952

set	nearest real	recon	u
band: held-out real	0.665	0.914	0.999
band: real + 0.5x noise	0.825	1.058	0.978
band: real + 1x noise	1.186	1.399	0.917

L2.attn → L2.ffn

set	nearest real	recon	u
sample $\alpha=0.3$	0.847	0.682	1.030
sample $\alpha=0.7$	0.895	0.745	1.023
sample $\alpha=1.0$	0.968	0.813	1.017
real held-out images	0.712	0.852	0.999
real + 1x noise images	1.165	1.258	0.935
band: held-out real	0.712	0.852	0.999
band: real + 0.5x noise	0.861	0.996	0.966
band: real + 1x noise	1.211	1.333	0.876

L3.attn → L3.ffn

set	nearest real	recon	u
sample $\alpha=0.3$	0.775	0.532	1.038
sample $\alpha=0.7$	0.841	0.595	1.032
sample $\alpha=1.0$	0.912	0.643	1.019
real held-out images	0.748	0.800	0.999
real + 1x noise images	1.162	1.171	0.914
band: held-out real	0.748	0.800	0.999
band: real + 0.5x noise	0.895	0.946	0.956
band: real + 1x noise	1.236	1.292	0.831

7. Decoder capacity and hyper-parameters (SimpleStories-5M, three locations)

Every setting scored by held-out reconstruction / spacing (final and best epoch on the validation split) and by the samples at $\alpha = 0.3$. default = hidden (512, 256), 300 epochs, lr 2e-3, batch 512, λ_{geom} 0.1.

embed — latent dimension (default architecture):

m	val recon (best)	train recon	u	nearest real	coverage	ESS
8	0.831	0.670	1.215	0.710	0.991	4898
16	0.788	0.563	1.149	0.642	0.986	1907
24	0.763	0.479	1.121	0.579	0.955	946
32	0.743	0.423	1.093	0.552	0.933	876
48	0.707	0.355	1.072	0.504	0.872	361
64	0.676	0.307	1.050	0.454	0.823	24.636

embed — architecture / training variants at m = 16 (sorted by held-out reconstruction):

variant	params	val recon	best (epoch)	train recon	u	coverage	ESS
hidden (512, 512, 256), 300 ep	1.06M	0.782	0.782 @ 299	0.506	1.120	0.955	1859
hidden (512, 256), 300 ep, {'lam_geom': 0.03}	0.53M	0.786	0.786 @ 275	0.555	1.144	0.984	1554
hidden (512, 256), 300 ep, {'lam_geom': 0.3}	0.53M	0.789	0.788 @ 250	0.572	1.157	0.984	2676
default	0.53M	0.788	0.788 @ 275	0.563	1.149	0.986	1907
hidden (2048, 1024), 300 ep	5.28M	0.790	0.790 @ 299	0.550	1.152	0.977	1468
hidden (512, 256), 600 ep	0.53M	0.799	0.792 @ 325	0.448	1.087	0.946	1240
hidden (512, 256), 300 ep, {'lr': 0.001}	0.53M	0.792	0.792 @ 299	0.593	1.168	0.987	2918
hidden (512, 256), 300 ep, {'lr': 0.005}	0.53M	0.792	0.792 @ 299	0.506	1.122	0.964	1097
hidden (256, 128), 300 ep	0.20M	0.795	0.794 @ 200	0.584	1.169	0.977	3392
hidden (512,), 300 ep	0.28M	0.795	0.795 @ 250	0.597	1.175	0.989	3249
hidden (1024, 512), 300 ep	1.59M	0.795	0.795 @ 299	0.556	1.149	0.981	723
hidden (512, 256), 300 ep, {'batch_size': 2048}	0.53M	0.799	0.799 @ 299	0.612	1.175	0.989	2766
hidden (512, 256), 300 ep, {'batch_size': 128}	0.53M	0.807	0.799 @ 125	0.540	1.148	0.963	1144
hidden (512, 256), 100 ep	0.53M	0.802	0.802 @ 99	0.630	1.200	0.990	4633

variant	params	val recon	best (epoch)	train recon	u	coverage	ESS
hidden (1024, 1024, 512), 300 ep	3.69M	0.822	0.822 @ 299	0.705	1.228	0.990	8523

L2.ffn — latent dimension (default architecture):

m	val recon (best)	train recon	u	nearest real	coverage	ESS
8	0.772	0.582	1.148	0.663	0.991	6063
16	0.725	0.492	1.096	0.611	0.990	3770
24	0.694	0.410	1.068	0.563	0.984	2363
32	0.662	0.356	1.051	0.537	0.968	1247
48	0.614	0.287	1.038	0.501	0.935	1429
64	0.573	0.239	1.028	0.475	0.901	1243

L2.ffn — architecture / training variants at m = 16 (sorted by held-out reconstruction):

variant	params	val recon	best (epoch)	train recon	u	coverage	ESS
hidden (1024, 1024, 512), 300 ep	3.69M	0.723	0.719 @ 200	0.397	1.052	0.974	1845
hidden (512, 512, 256), 300 ep	1.06M	0.723	0.721 @ 200	0.427	1.068	0.983	2903
hidden (512, 256), 300 ep, {'lam_geom': 0.03}	0.53M	0.722	0.722 @ 299	0.481	1.094	0.989	3851
hidden (256, 128), 300 ep	0.20M	0.723	0.722 @ 250	0.507	1.116	0.991	4110
default	0.53M	0.725	0.725 @ 275	0.492	1.096	0.990	3770
hidden (1024, 512), 300 ep	1.59M	0.726	0.726 @ 275	0.486	1.093	0.989	2379
hidden (512, 256), 300 ep, {'lr': 0.001}	0.53M	0.727	0.726 @ 275	0.517	1.112	0.991	3351
hidden (512, 256), 300 ep, {'lr': 0.005}	0.53M	0.728	0.728 @ 299	0.439	1.081	0.988	3554
hidden (512, 256), 600 ep	0.53M	0.736	0.728 @ 300	0.392	1.055	0.984	2590
hidden (512, 256), 300 ep, {'lam_geom': 0.3}	0.53M	0.729	0.729 @ 299	0.496	1.103	0.989	3970
hidden (512,), 300 ep	0.28M	0.730	0.729 @ 225	0.524	1.117	0.990	2823
hidden (512, 256), 300 ep, {'batch_size': 2048}	0.53M	0.731	0.731 @ 225	0.536	1.125	0.991	3411
hidden (512, 256), 100 ep	0.53M	0.735	0.735 @ 99	0.552	1.150	0.991	4657
hidden (512, 256), 300 ep, {'batch_size': 128}	0.53M	0.738	0.735 @ 150	0.468	1.099	0.986	4011
hidden (2048, 1024), 300 ep	5.28M	0.751	0.751 @ 299	0.588	1.157	0.989	7192

L5.ffn — latent dimension (default architecture):

m	val recon (best)	train recon	u	nearest real	coverage	ESS
8	0.767	0.523	1.165	0.694	0.995	6780
16	0.704	0.411	1.114	0.620	0.991	3730
24	0.651	0.329	1.075	0.559	0.986	2288
32	0.606	0.275	1.055	0.523	0.970	2170
48	0.542	0.211	1.038	0.491	0.925	2084
64	0.491	0.171	1.025	0.458	0.809	1049

L5.ffn — architecture / training variants at m = 16 (sorted by held-out reconstruction):

variant	params	val recon	best (epoch)	train recon	u	coverage	ESS
hidden (512, 512, 256), 300 ep	1.06M	0.702	0.701 @ 225	0.366	1.085	0.992	2843
hidden (512, 256), 300 ep, {'lr': 0.005}	0.53M	0.702	0.701 @ 250	0.376	1.095	0.992	3926
hidden (512, 256), 600 ep	0.53M	0.702	0.701 @ 525	0.337	1.063	0.989	1442
hidden (512, 256), 300 ep, {'lam_geom': 0.03}	0.53M	0.704	0.704 @ 299	0.404	1.114	0.993	4860
default	0.53M	0.704	0.704 @ 250	0.411	1.114	0.991	3730
hidden (512, 256), 300 ep, {'lam_geom': 0.3}	0.53M	0.706	0.705 @ 250	0.416	1.113	0.991	4357
hidden (1024, 512), 300 ep	1.59M	0.706	0.706 @ 275	0.410	1.115	0.992	3417
hidden (512, 256), 300 ep, {'batch_size': 128}	0.53M	0.708	0.708 @ 299	0.391	1.104	0.992	3322
hidden (512, 256), 300 ep, {'lr': 0.001}	0.53M	0.708	0.708 @ 275	0.432	1.124	0.992	3882
hidden (256, 128), 300 ep	0.20M	0.709	0.709 @ 299	0.425	1.131	0.991	4370
hidden (512,), 300 ep	0.28M	0.716	0.716 @ 250	0.438	1.126	0.993	4706
hidden (512, 256), 300 ep, {'batch_size': 2048}	0.53M	0.718	0.718 @ 275	0.454	1.139	0.991	4302
hidden (512, 256), 100 ep	0.53M	0.721	0.721 @ 99	0.465	1.155	0.992	5954
hidden (2048, 1024), 300 ep	5.28M	0.725	0.725 @ 275	0.474	1.150	0.991	6530
hidden (1024, 1024, 512), 300 ep	3.69M	0.825	0.825 @ 299	0.648	1.223	0.981	5117

8. How the samples look to the model (5M_downstream, SimpleStories/SimpleStories-5M)

Embedding samples are placed at the last position of the prefix and the *full* model is run. (a) Next-token distributions: Jensen-Shannon divergence to the distribution of the sample's nearest real token, second-nearest, and a random token; entropy; top-1 agreement. (b) Layer-by-layer trajectory of the images against every location's own manifold and bands, with perp = distance from the segment joining the images of the two nearest real anchors (in spacings) and the fraction of images whose projection falls between those two anchor images.

set	JS to nearest token	JS to 2nd nearest	JS to random token	entropy (real: 2.86)	top-1 = nearest	top-1 = random
held-out real tokens	0.457	0.515	0.606	2.800	0.193	0.038
samples $\alpha=0.3$	0.212	0.291	0.565	3.380	0.316	0.064
samples $\alpha=0.7$	0.274	0.327	0.554	3.272	0.260	0.067
samples $\alpha=1.5$	0.368	0.396	0.542	2.858	0.216	0.071
real + 1x noise	0.508	0.522	0.575	3.883	0.076	0.043

Examples (α = first sweep value): the two nearest real tokens of a sample, the sample's top-3 next-token predictions, and the nearest token's top-3.

nearest	2nd nearest	sample predicts	nearest token predicts
himself	herself	. , and	. , and
guiding	bringing	them him away	star light place
##elt	pizz	##a ##er ##ous	##a ##pt ##er
jose	leo	who . ,	and . ,
cool	cold	and , -	, ##er dinosaur
clouds	snowflakes	. , that	that of .

kernel score u of the images, per location:

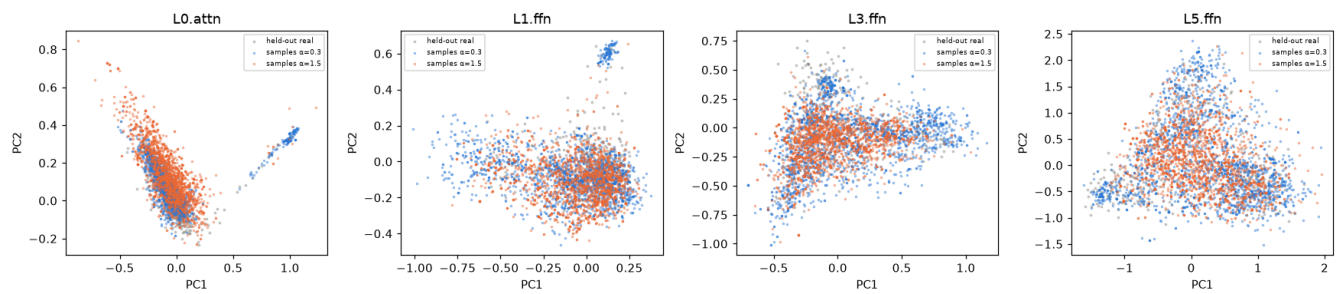
location	samples $\alpha=0.3$	samples $\alpha=0.7$	samples $\alpha=1.5$	held-out real	real + 1x noise	band: real + 1x noise
embed	1.149	1.088	0.758	0.995	0.707	0.707
L0.attn	1.144	1.084	0.768	0.994	0.707	0.721
L0.ffn	1.035	1.008	0.795	1.001	0.781	0.695
L1.attn	1.021	0.997	0.796	1.000	0.783	0.694
L1.ffn	1.000	0.984	0.842	1.000	0.847	0.690
L2.attn	0.998	0.983	0.848	1.000	0.847	0.690
L2.ffn	0.996	0.989	0.903	0.998	0.891	0.691
L3.attn	0.996	0.988	0.903	0.999	0.902	0.692
L3.ffn	0.997	0.995	0.951	0.999	0.942	0.689
L4.attn	0.998	0.994	0.943	0.998	0.946	0.685
L4.ffn	0.999	0.998	0.968	0.997	0.971	0.686
L5.attn	0.999	0.997	0.965	0.998	0.972	0.686
L5.ffn	0.995	0.993	0.974	0.998	0.995	0.697

nearest real / spacing of the images, per location:

location	samples $\alpha=0.3$	samples $\alpha=0.7$	samples $\alpha=1.5$	held-out real	real + 1x noise	band: real + 1x noise
embed	0.642	0.737	1.150	0.948	1.368	1.369
L0.attn	0.655	0.748	1.162	0.965	1.388	1.379
L0.ffn	0.741	0.808	1.134	0.902	1.249	1.320
L1.attn	0.759	0.819	1.135	0.899	1.247	1.314
L1.ffn	0.796	0.849	1.113	0.911	1.203	1.308
L2.attn	0.799	0.851	1.116	0.910	1.205	1.307
L2.ffn	0.813	0.854	1.083	0.895	1.162	1.309
L3.attn	0.809	0.858	1.084	0.897	1.152	1.303
L3.ffn	0.814	0.863	1.056	0.904	1.103	1.301
L4.attn	0.817	0.873	1.065	0.899	1.110	1.309
L4.ffn	0.822	0.881	1.050	0.913	1.095	1.307
L5.attn	0.825	0.887	1.062	0.919	1.087	1.314
L5.ffn	0.858	0.911	1.053	0.920	1.051	1.325

perp. distance from the anchor-image segment / spacing of the images, per location:

location	samples $\alpha=0.3$	samples $\alpha=0.7$	samples $\alpha=1.5$	held-out real	real + 1x noise
embed	0.517	0.620	1.063	0.826	1.277
L0.attn	0.532	0.632	1.073	0.840	1.300
L0.ffn	0.653	0.712	1.056	0.808	1.186
L1.attn	0.676	0.731	1.061	0.813	1.186
L1.ffn	0.718	0.764	1.046	0.825	1.165
L2.attn	0.722	0.767	1.050	0.829	1.164
L2.ffn	0.740	0.784	1.022	0.851	1.150
L3.attn	0.740	0.787	1.027	0.848	1.145
L3.ffn	0.751	0.794	1.005	0.862	1.122
L4.attn	0.756	0.799	1.012	0.857	1.123
L4.ffn	0.763	0.818	1.018	0.907	1.121
L5.attn	0.768	0.829	1.037	0.907	1.146
L5.ffn	0.810	0.870	1.054	0.950	1.140



PCA of the real states (grey) and of the images of samples at the smallest (blue) and largest (orange) α , at four locations.